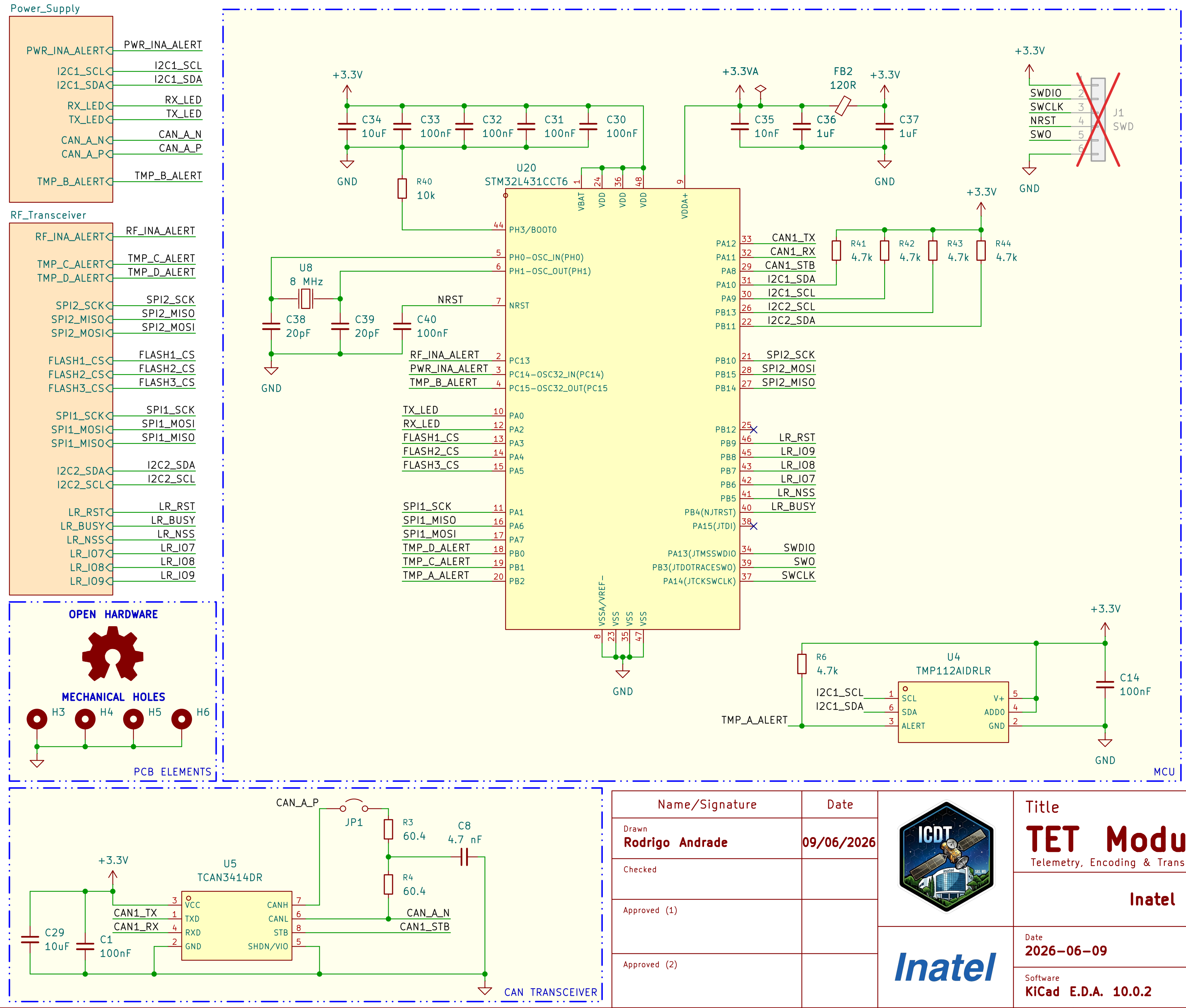

MCU

– Main processing unit based on the STM32L431 microcontroller.

– Responsible for telemetry acquisition, packet encoding, peripheral control, communication interfaces, and RF subsystem management.

– Interfaces with external memories, sensors, CAN bus, and LoRa transceiver through SPI, I2C, and GPIO peripherals.

CAN

– CAN bus physical interface implemented using the TCAN3414 transceiver.

– Converts MCU logic-level CAN signals into differential CANH/CANL bus signals compatible with external spacecraft subsystems.

SWD

– Standard ARM SWD programming and debugging interface.

– Provides firmware upload, debugging access, and real-time MCU inspection through SWDIO, SWCLK, NRST, and SWO signals.

CLOCK

– External 8 MHz oscillator used as the primary high-speed clock reference for the STM32 MCU.

– Provides accurate timing for communication peripherals and RF-related operations.

TEMPERATURE

– Local board temperature monitoring using the TMP112 digital temperature sensor connected via I2C bus.

– Generates alert interrupts when programmable temperature thresholds are exceeded.

	Name/Signature	Date
Drawn	Rodrigo Andrade	09/06/2026
Checked		
Approved (1)		
Approved (2)		

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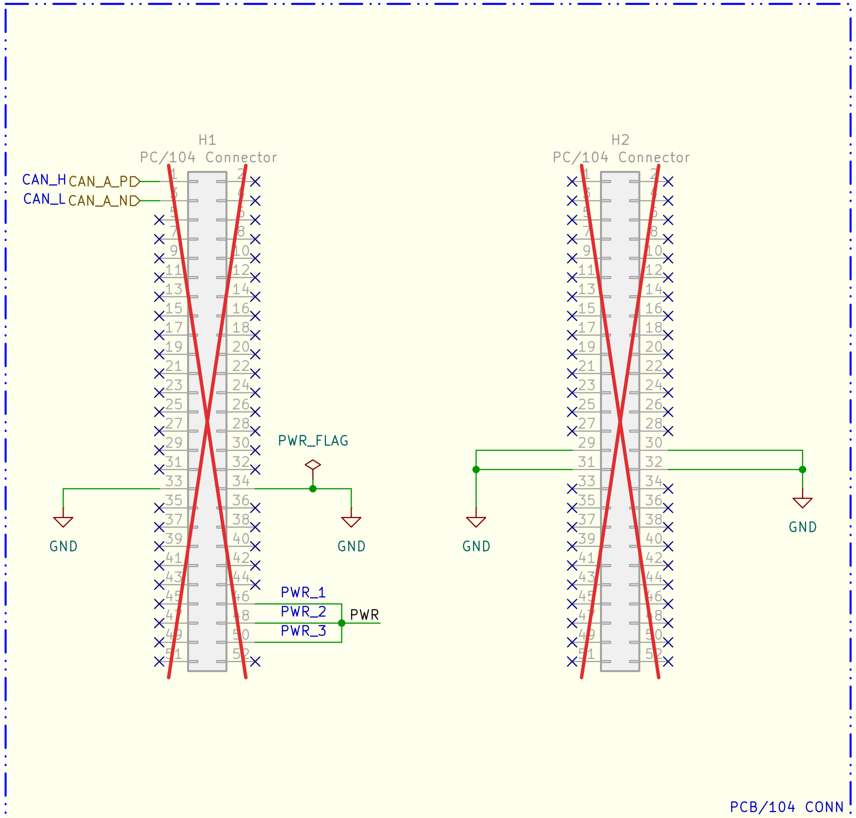
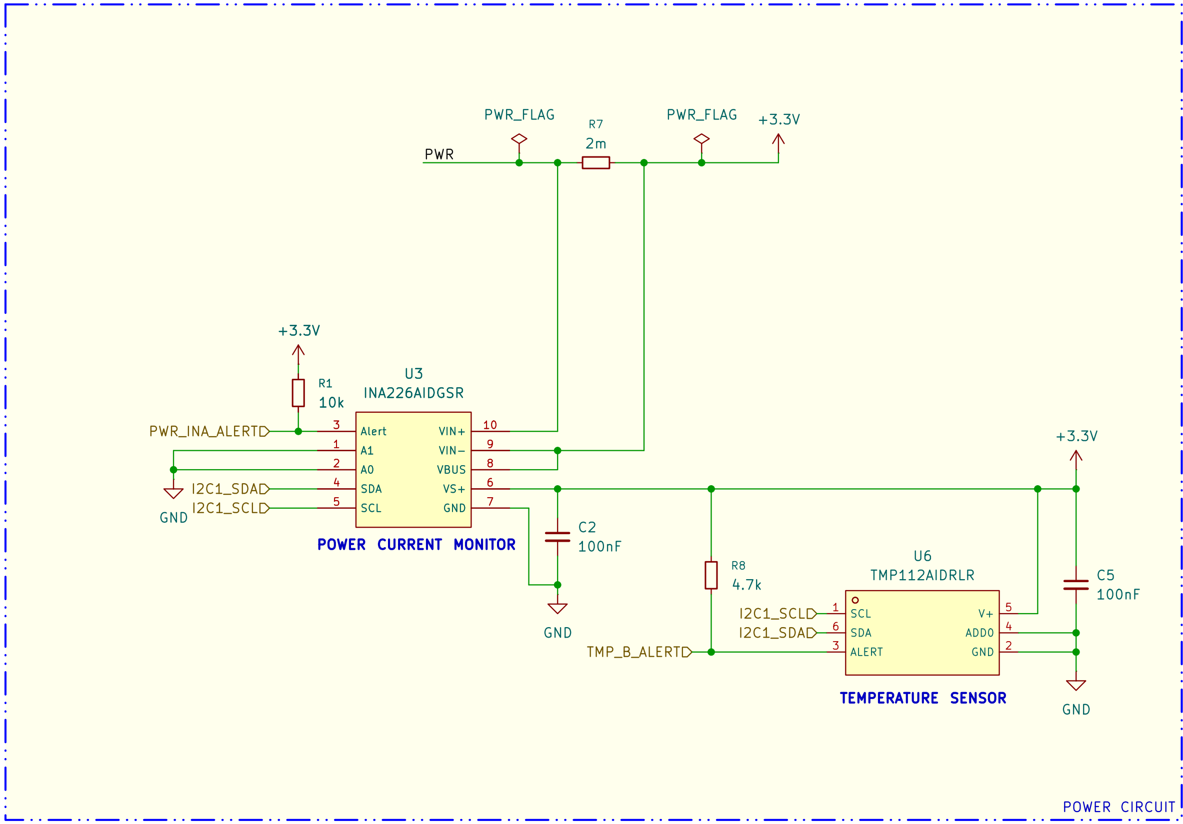
Revised · To be revised · Needs fix

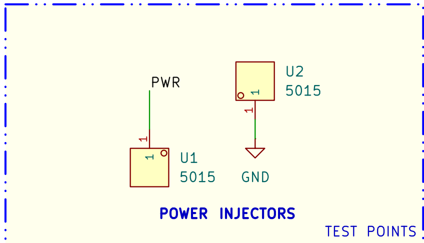
Title
TET Module
Telemetry, Encoding & Transmission Module

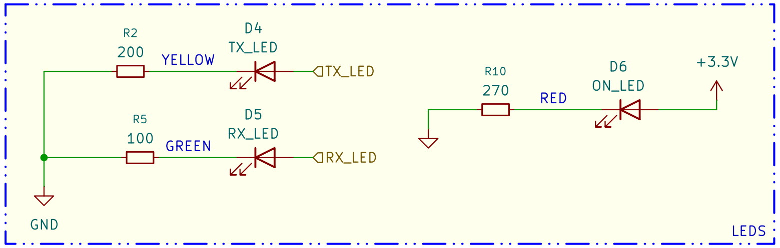
Inatel CubeSat Design Team

Date	Revision	Size
2026-06-09	v1r1	A4
Software	**Sheet**	**Code**
KiCad E.D.A. 10.0.2	1/3	TET-v1r1









POWER SUPPLY

- Main power conditioning stage of the module.
- Receives regulated 5 V input from the EPS through the PC/104 connector and distributes power to the onboard subsystems.
- Schottky diode OR−ing network used to combine multiple EPS power channels (PWR_1, PWR_2, PWR_3).
- Prevents reverse current flow between channels and increases power redundancy.
- High−side current and voltage monitoring circuit based on the INA226 sensor.
- Measures input current consumption and bus voltage from the EPS power line through a low−ohmic shunt resistor.
- Low−dropout voltage regulator generating the main 3.3 V rail used by the digital and RF subsystems.
- Converts the incoming 5 V supply into a regulated low−noise output voltage.
- Monitors the temperature of the power regulation section and surrounding circuitry.
- Connected through the primary I2C bus and capable of generating thermal alert interrupts.
- Multiple PC/104 connector pins used for redundant power injection into the module from the spacecraft EPS.
- Visual indicators used for power presence and communication activity monitoring.
- Includes power−on, TX, and RX status indication.

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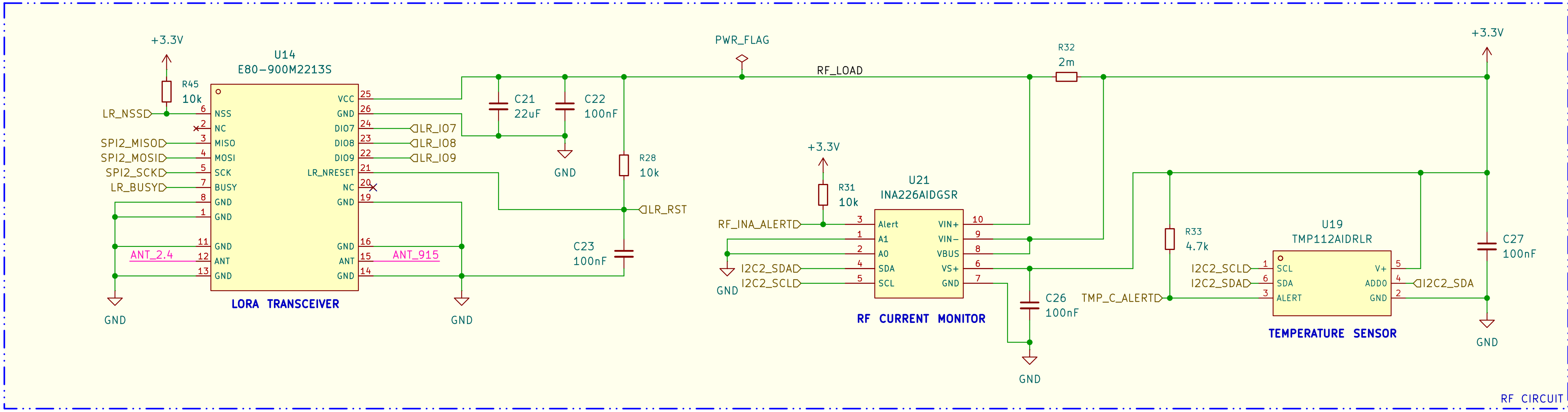
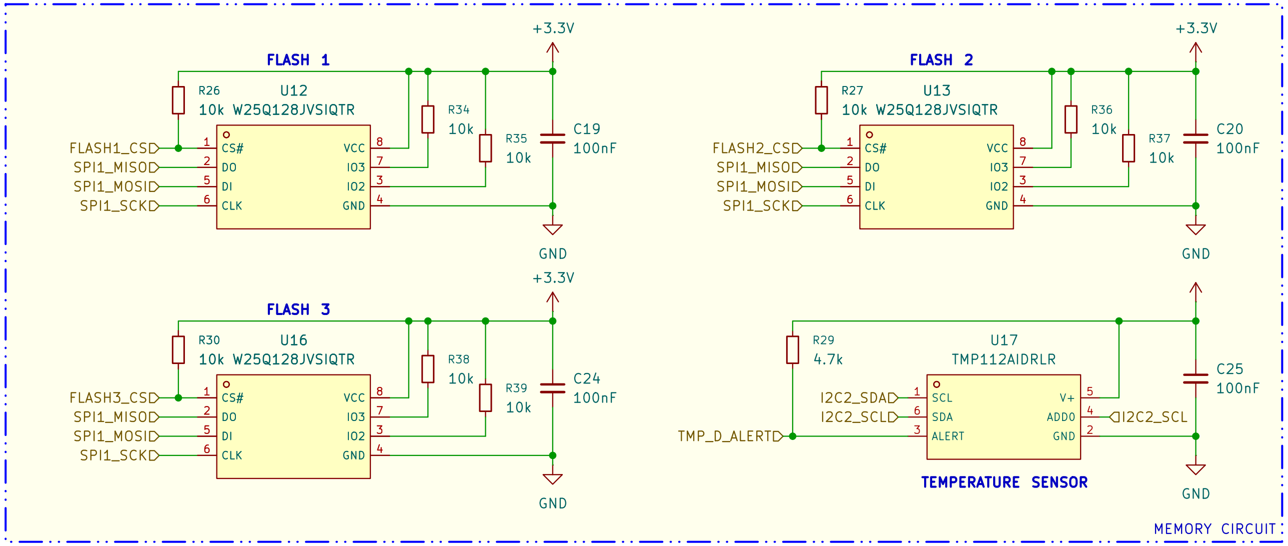
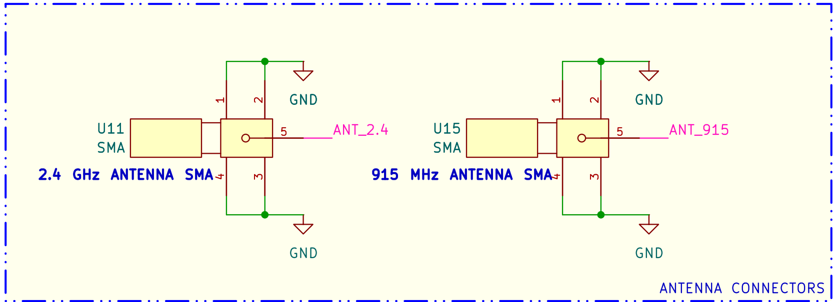
Software	**Sheet**	**Code**
KiCad E.D.A. 10.0.2	2/3	TET−v1r1





RF Transceiver

- Dual-band LoRa transceiver supporting 915 MHz and 2.4 GHz operation.
- Based on the Semtech LR1121 RF transceiver architecture.
- Supports LoRa, GFSK, FLRC, and LR-FHSS modulation schemes.
- Dedicated SMA RF outputs for 915 MHz and 2.4 GHz external antennas.
- SPI interface used for RF configuration and packet communication.
- GPIO control lines used for reset, busy status, and interrupt signaling.
- Designed for long-range telemetry and low-power wireless communication.
- RF subsystem power consumption monitored through an INA226 high-side current monitor.
- Local RF temperature monitoring implemented using TMP112 digital temperature sensors.
- External SPI flash memories used for telemetry buffering and RF data storage.
- RF section powered through an independent monitored 3.3 V rail.



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2026-06-09	v1r1	A4
Software	**Sheet**	**Code**
KiCad E.D.A. 10.0.2	3/3	TET-v1r1